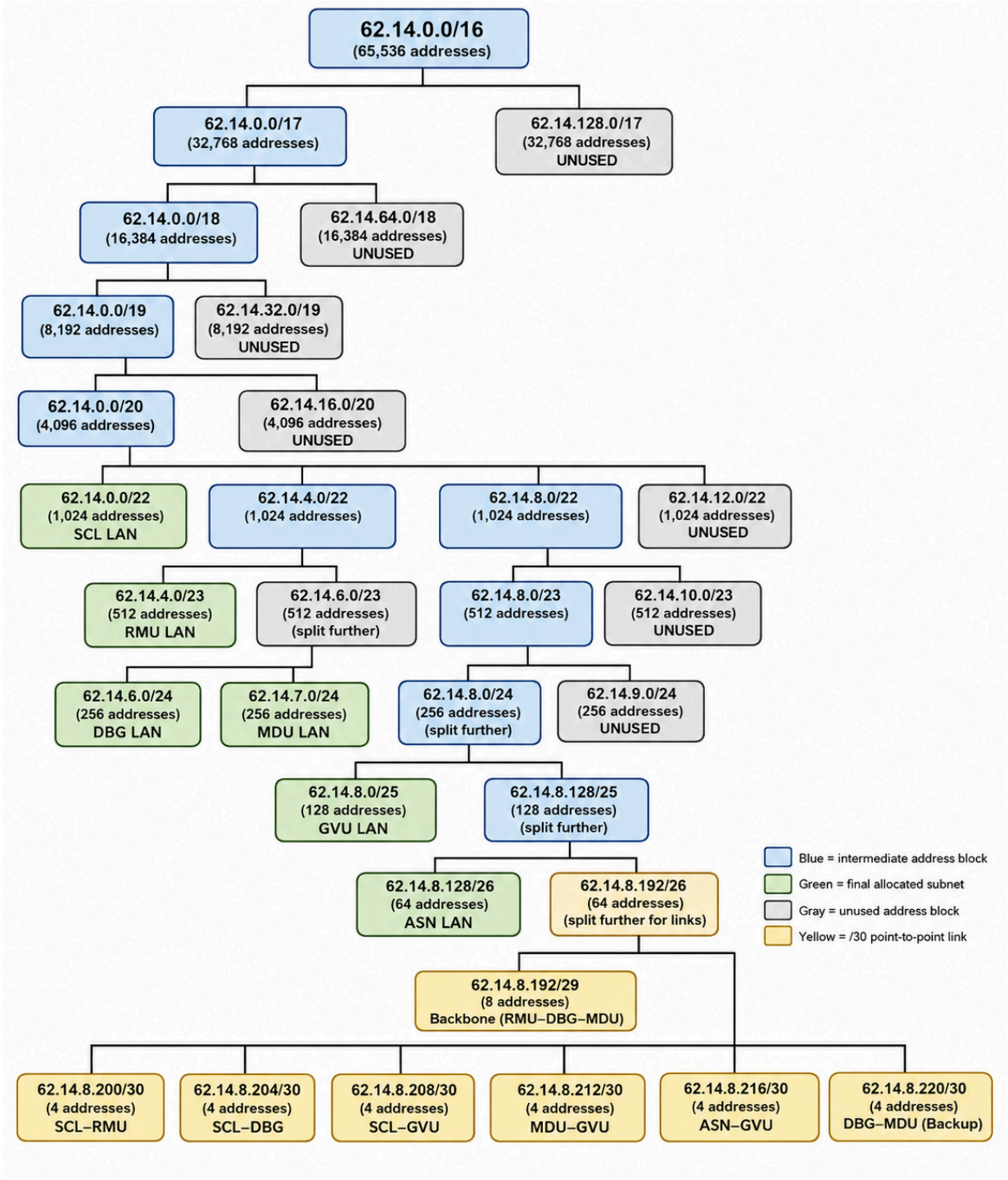
# 1)VLSM TREE-




Subnets carved from 62.14.0.0/16, largest requirement first:

| Segment | Hosts needed | Prefix | Subnet | Usable range |
|---|---|---|---|---|
| SCL LAN | 520 | /22 | 62.14.0.0/22 | 62.14.0.1 – 62.14.3.254 |
| RMU LAN | 300 | /23 | 62.14.4.0/23 | 62.14.4.1 – 62.14.5.254 |
| DBG LAN | 220 | /24 | 62.14.6.0/24 | 62.14.6.1 – 62.14.6.254 |

| | | | | |
|---|---|---|---|---|
| MDU LAN | 160 | /24 | 62.14.7.0/24 | 62.14.7.1 – 62.14.7.254 |
| GVU LAN | 90 | /25 | 62.14.8.0/25 | 62.14.8.1 – 62.14.8.126 |
| ASN LAN | 45 | /26 | 62.14.8.128/26 | 62.14.8.129 – 62.14.8.190 |
| Backbone (RMU/DBG/MDU) | 3 routers | /29 | 62.14.8.192/29 | 62.14.8.193 – 62.14.8.198 |
| SCL–RMU link | 2 | /30 | 62.14.8.200/30 | .201 – .202 |
| SCL–DBG link | 2 | /30 | 62.14.8.204/30 | .205 – .206 |
| SCL–GVU link | 2 | /30 | 62.14.8.208/30 | .209 – .210 |
| MDU–GVU link | 2 | /30 | 62.14.8.212/30 | .213 – .214 |
| ASN–GVU link | 2 | /30 | 62.14.8.216/30 | .217 – .218 |
| DBG–MDU backup link | 2 | /30 | 62.14.8.220/30 | .221 – .222 |

# 2.IP Addressing Table

## Router interfaces

| Router | Interface | IP Address | Subnet Mask | Connects to |
|---|---|---|---|---|
| SCL | Gi0/0 | 62.14.0.1 | 255.255.252.0 | SCL-SW (LAN) |
| SCL | Gi0/1 | 62.14.8.201 | 255.255.255.252 | RMU |
| SCL | Se0/3/0 | 62.14.8.205 | 255.255.255.252 | DBG |
| SCL | Se0/3/1 | 62.14.8.209 | 255.255.255.252 | GVU |
| RMU | Gi0/0 | 62.14.8.202 | 255.255.255.252 | SCL |
| RMU | Gi0/1 | 62.14.4.1 | 255.255.254.0 | RMU-SW (LAN) |
| RMU | Gi0/2 | 62.14.8.193 | 255.255.255.248 | Backbone switch |
| DBG | Gi0/0 | 62.14.6.1 | 255.255.255.0 | DBG-SW (LAN) |
| DBG | Gi0/1 | 62.14.8.194 | 255.255.255.248 | Backbone switch |
| DBG | Se0/0/0 | 62.14.8.206 | 255.255.255.252 | SCL |
| DBG | Se0/0/1 | 62.14.8.221 | 255.255.255.252 | MDU (backup link) |
| MDU | Gi0/0 | 62.14.7.1 | 255.255.255.0 | MDU-SW (LAN) |
| MDU | Gi0/1 | 62.14.8.195 | 255.255.255.248 | Backbone switch |
| MDU | Se0/0/0 | 62.14.8.222 | 255.255.255.252 | DBG (backup link) |
| MDU | Se0/0/1 | 62.14.8.213 | 255.255.255.252 | GVU |
| GVU | Gi0/0 | 62.14.8.1 | 255.255.255.128 | GVU-SW (LAN) |
| GVU | Gi0/1 | 62.14.8.218 | 255.255.255.252 | ASN |
| GVU | Se0/0/0 | 62.14.8.210 | 255.255.255.252 | SCL |
| GVU | Se0/0/1 | 62.14.8.214 | 255.255.255.252 | MDU |
| ASN | Gi0/0 | 62.14.8.129 | 255.255.255.192 | ASN-SW (LAN) |
| ASN | Gi0/1 | 62.14.8.217 | 255.255.255.252 | GVU |

## Servers and printers (static)

| Device | IP Address | Subnet Mask | Gateway | DNS Server |
|---|---|---|---|---|
| SCL-DNS | 62.14.0.2 | 255.255.252.0 | 62.14.0.1 | 62.14.0.2 |
| SCL-WEB | 62.14.0.3 | 255.255.252.0 | 62.14.0.1 | 62.14.0.2 |

| | | | | |
|---|---|---|---|---|
| SCL-DHCP | 62.14.0.4 | 255.255.252.0 | 62.14.0.1 | 62.14.0.2 |
| SCL-EMAIL | 62.14.0.5 | 255.255.252.0 | 62.14.0.1 | 62.14.0.2 |
| SCL-PRINTER | 62.14.0.6 | 255.255.252.0 | 62.14.0.1 | 62.14.0.2 |
| RMU-PRINTER | 62.14.4.2 | 255.255.254.0 | 62.14.4.1 | 62.14.0.2 |
| DBG-PRINTER | 62.14.6.2 | 255.255.255.0 | 62.14.6.1 | 62.14.0.2 |
| MDU-EMAIL | 62.14.7.2 | 255.255.255.0 | 62.14.7.1 | 62.14.0.2 |
| MDU-PRINTER | 62.14.7.3 | 255.255.255.0 | 62.14.7.1 | 62.14.0.2 |
| GVU-PRINTER | 62.14.8.2 | 255.255.255.128 | 62.14.8.1 | 62.14.0.2 |
| MDU-DHCP | 62.14.7.4 | 255.255.255.0 | 62.14.7.1 | 62.14.0.2 |
| ASN-PRINTER | 62.14.8.130 | 255.255.255.192 | 62.14.8.129 | 62.14.0.2 |
| ASN-PC1 | 62.14.8.131 | 255.255.255.192 | 62.14.8.129 | 62.14.0.2 |
| ASN-PC2 | 62.14.8.132 | 255.255.255.192 | 62.14.8.129 | 62.14.0.2 |

## DHCP pools

| Pool | Type | Network | Gateway | DNS | Start address |
|---|---|---|---|---|---|
| SCL_POOL | Server (SCL-DHCP) | 62.14.0.0/22 | 62.14.0.1 | 62.14.0.2 | 62.14.0.7 |
| GVU_POOL | Server (SCL-DHCP, via relay) | 62.14.8.0/25 | 62.14.8.1 | 62.14.0.2 | 62.14.8.10 |
| MDU_POOL | Server (MDU-DHCP) | 62.14.7.0/24 | 62.14.7.1 | 62.14.0.2 | 62.14.7.5 |
| RMU_POOL | Router-based (RMU) | 62.14.4.0/23 | 62.14.4.1 | 62.14.0.2 | 62.14.4.62 (Member 1 offset) |
| DBG_POOL | Router-based (DBG) | 62.14.6.0/24 | 62.14.6.1 | 62.14.0.2 | 62.14.6.14 (Member 2 offset) |

# 3.Router Configuration Commands

## SCL ROUTER CONFIGURATION:

Router#show running-config
Building configuration...
Current configuration : 1460 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
no ip cef
no ipv6 cef

```
license udi pid CISCO2911/K9 sn FTX152468V3-
spanning-tree mode pvst
interface GigabitEthernet0/0
ip address 62.14.0.1 255.255.252.0
ip helper-address 62.14.0.4
duplex auto
speed auto
interface GigabitEthernet0/1
ip address 62.14.8.201 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/0/1
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Serial0/2/0
no ip address
clock rate 2000000
!
interface Serial0/2/1
no ip address
```

```
clock rate 2000000
!
interface Serial0/3/0
ip address 62.14.8.205 255.255.255.252
clock rate 2000000
!
interface Serial0/3/1
ip address 62.14.8.209 255.255.255.252
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
passive-interface GigabitEthernet0/0
network 62.0.0.0
no auto-summary
!
ip classless
ip route 62.14.8.128 255.255.255.192 62.14.8.210
!
ip flow-export version 9
!
!
!

line con 0
!
line aux 0
!
line vty 0 4
login
!
!
end
```

# RMU ROUTER CONFIGURATION

```
Router#show running-config
Building configuration...
Current configuration : 1026 bytes
!
```

```
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
ip dhcp excluded-address 62.14.4.1 62.14.4.61
!
ip dhcp pool RMU_POOL
network 62.14.4.0 255.255.254.0
default-router 62.14.4.1
dns-server 62.14.0.2
!
!
!
ip cef
no ipv6 cef
!

!
license udi pid CISCO2911/K9 sn FTX1524U683-

!
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0
ip address 62.14.8.202 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 62.14.4.1 255.255.254.0
duplex auto
speed auto
!
interface GigabitEthernet0/2
```

```
ip address 62.14.8.193 255.255.255.248
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
passive-interface GigabitEthernet0/1
network 62.0.0.0
no auto-summary
!
ip classless
ip route 62.14.7.0 255.255.255.0 62.14.8.195
!
ip flow-export version 9
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

# DBG ROUTER CONFIGURATION:

```
Router#show running-config
Building configuration...
Current configuration : 1579 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
!
hostname Router
!
!
ip dhcp excluded-address 62.14.6.1 62.14.6.13
!
ip dhcp pool DBG_POOL
network 62.14.6.0 255.255.255.0
default-router 62.14.6.1
dns-server 62.14.0.2
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX15241R0B-
!
!
!
!
spanning-tree mode pvst
!
!
!
interface GigabitEthernet0/0
ip address 62.14.6.1 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 62.14.8.194 255.255.255.248
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
```

```
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
ip address 62.14.8.206 255.255.255.252
!
interface Serial0/0/1
ip address 62.14.8.221 255.255.255.252
clock rate 2000000
!
interface Serial0/1/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Serial0/2/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/2/1
no ip address
clock rate 2000000
shutdown
!
interface Serial0/3/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
```

```
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
passive-interface GigabitEthernet0/0
network 62.0.0.0
no auto-summary
!
ip classless
ip route 62.14.0.0 255.255.252.0 62.14.8.222 123
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

# MDU ROUTER CONFIGURATION:

```
Router#show running-config
Building configuration...
Current configuration : 1371 bytes
!
version 15.1
no service timestamps log datetime msec
```

```
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router

!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX15244FV3-
!
!
spanning-tree mode pvst

!
!
interface GigabitEthernet0/0
ip address 62.14.7.1 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 62.14.8.195 255.255.255.248
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
!
interface Serial0/0/0
ip address 62.14.8.222 255.255.255.252
!
interface Serial0/0/1
```

```
ip address 62.14.8.213 255.255.255.252
clock rate 2000000
!
interface Serial0/1/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Serial0/2/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/2/1
no ip address
clock rate 2000000
shutdown
!
interface Serial0/3/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
passive-interface GigabitEthernet0/0
```

```
network 62.0.0.0
no auto-summary
!
ip classless
!
ip flow-export version 9

line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
End
```

# GVU ROUTER CONFIGURATION:

```
Router#show running-config
Building configuration...
Current configuration : 1394 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
no ip cef
no ipv6 cef
!
license udi pid CISCO2911/K9 sn FTX15245956-
!
!
spanning-tree mode pvst
!
```

```
!
interface GigabitEthernet0/0
ip address 62.14.8.1 255.255.255.128
ip helper-address 62.14.0.4
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 62.14.8.218 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
!
interface Serial0/0/0
ip address 62.14.8.210 255.255.255.252
!
interface Serial0/0/1
ip address 62.14.8.214 255.255.255.252
!
interface Serial0/1/0
no ip address
clock rate 2000000
!
interface Serial0/1/1
no ip address
clock rate 2000000
!
interface Serial0/2/0
no ip address
clock rate 2000000
!
interface Serial0/2/1
no ip address
clock rate 2000000
!
interface Serial0/3/0
```

```
no ip address
clock rate 2000000
!
interface Serial0/3/1
no ip address
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
redistribute static
passive-interface GigabitEthernet0/0
network 62.0.0.0
no auto-summary
!
ip classless
ip route 62.14.8.128 255.255.255.192 62.14.8.217
!
ip flow-export version 9
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

# ASN ROUTER CONFIGURATION:

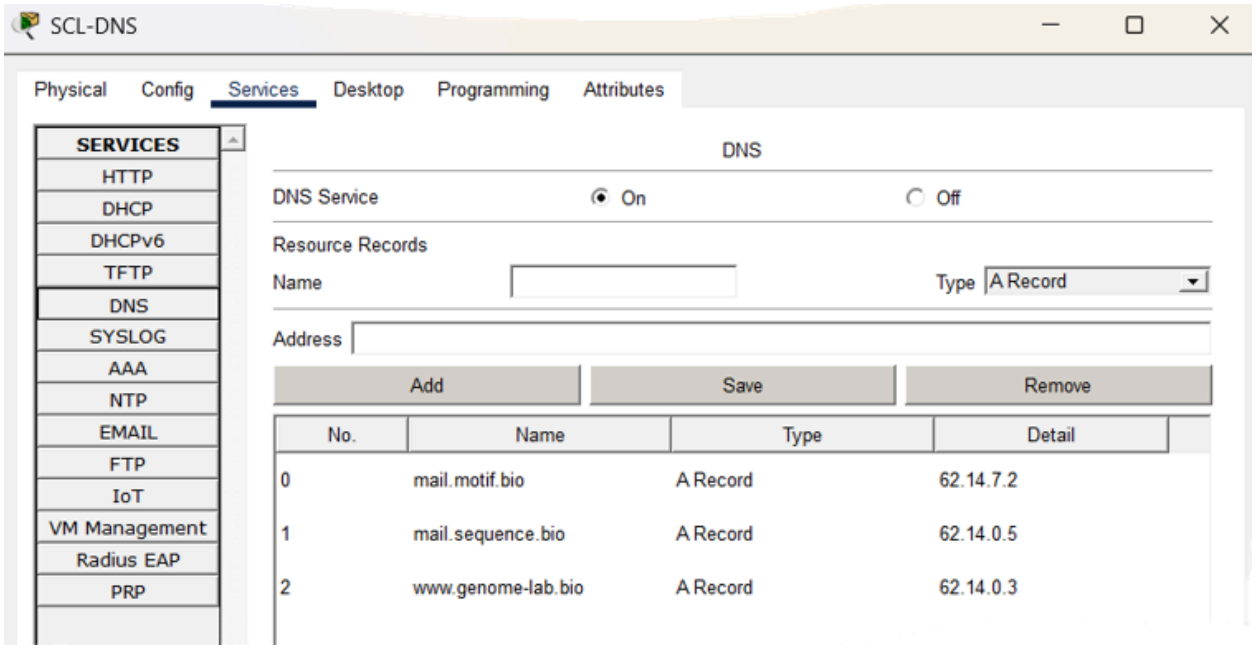
```
Router#show running-config
Building configuration...
Current configuration : 1035 bytes
!
```

```
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524BUB8-
!
!
spanning-tree mode pvst
!
!
!
interface GigabitEthernet0/0
ip address 62.14.8.129 255.255.255.192
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 62.14.8.217 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
```

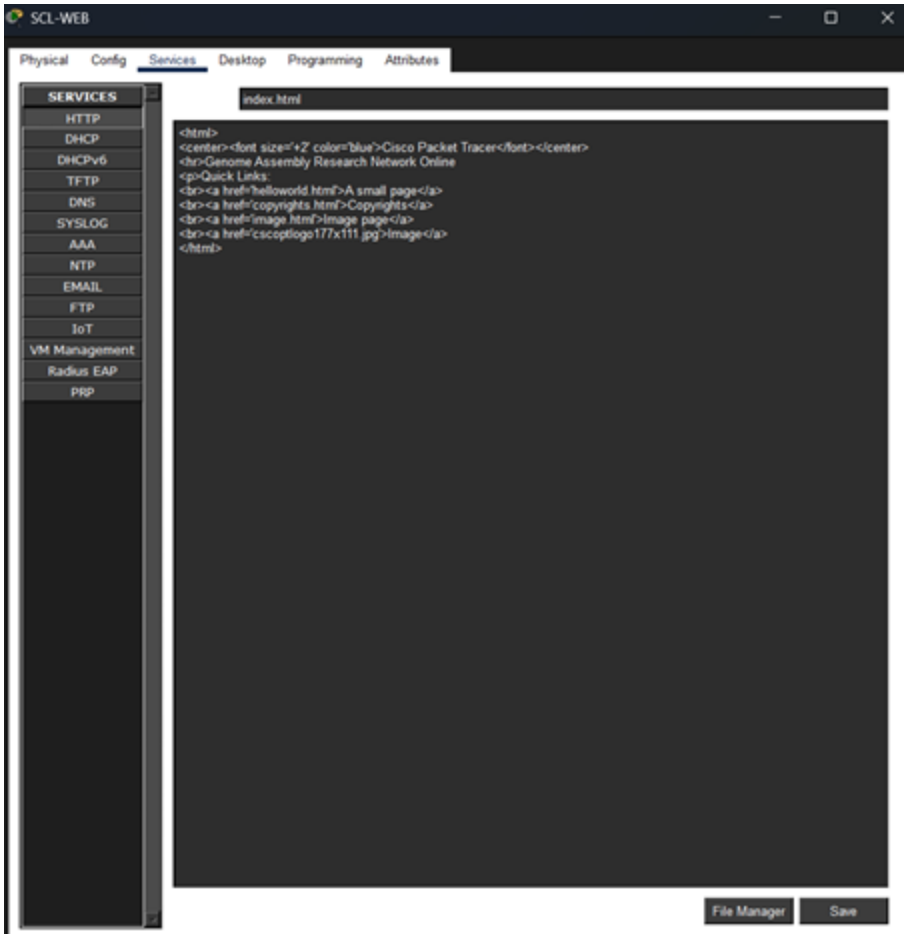
```
shutdown
!
ip classless
ip route 62.14.0.0 255.255.252.0 GigabitEthernet0/1
ip route 62.14.4.0 255.255.254.0 GigabitEthernet0/1
ip route 62.14.6.0 255.255.255.0 GigabitEthernet0/1
ip route 62.14.7.0 255.255.255.0 GigabitEthernet0/1
ip route 62.14.8.0 255.255.255.128 GigabitEthernet0/1
ip route 0.0.0.0 0.0.0.0 GigabitEthernet0/1
!
ip flow-export version 9
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

# 4.Server Configuration Screenshots

## DNS (SCL-DNS)-



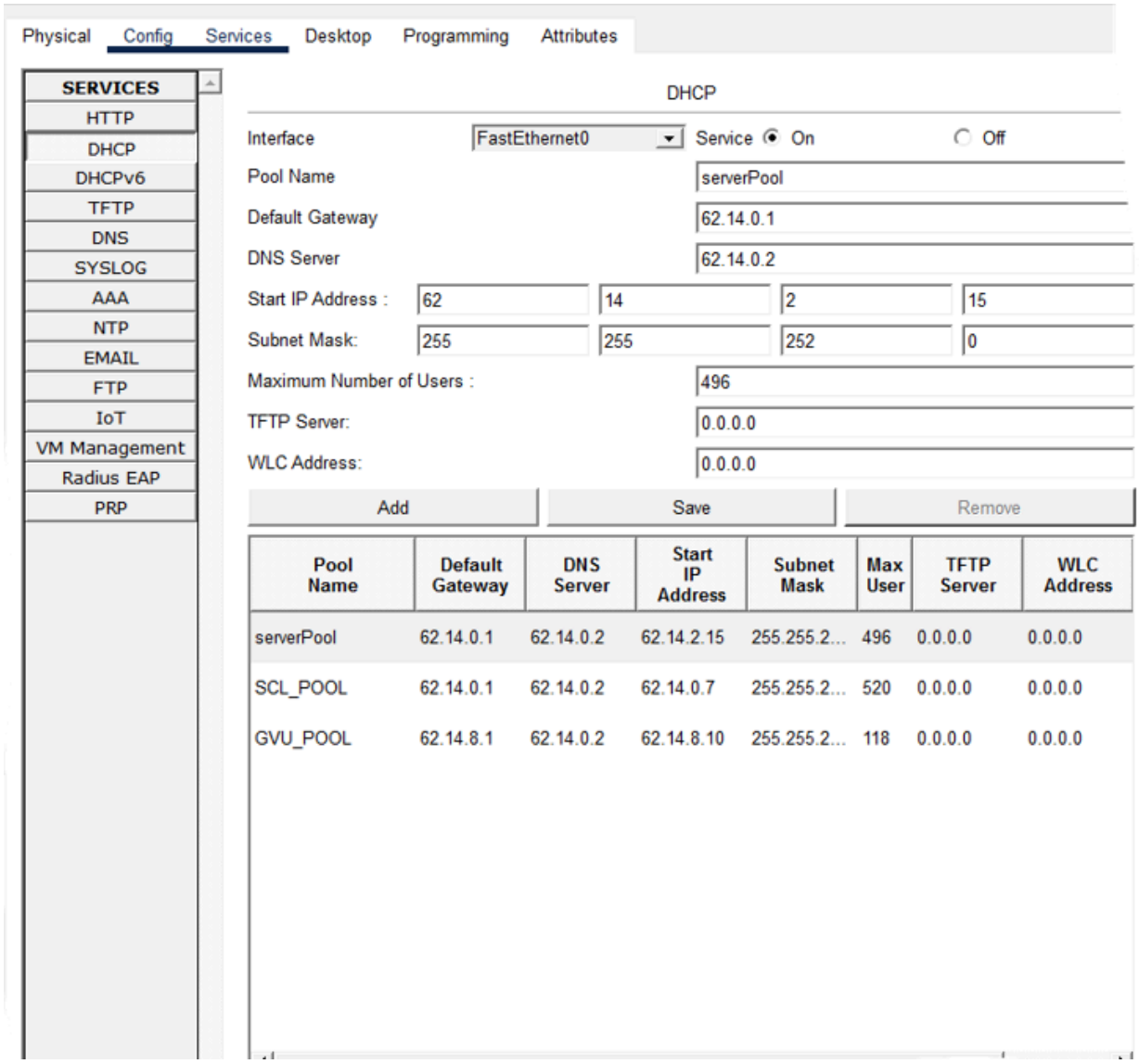

## Web (SCL-WEB)-



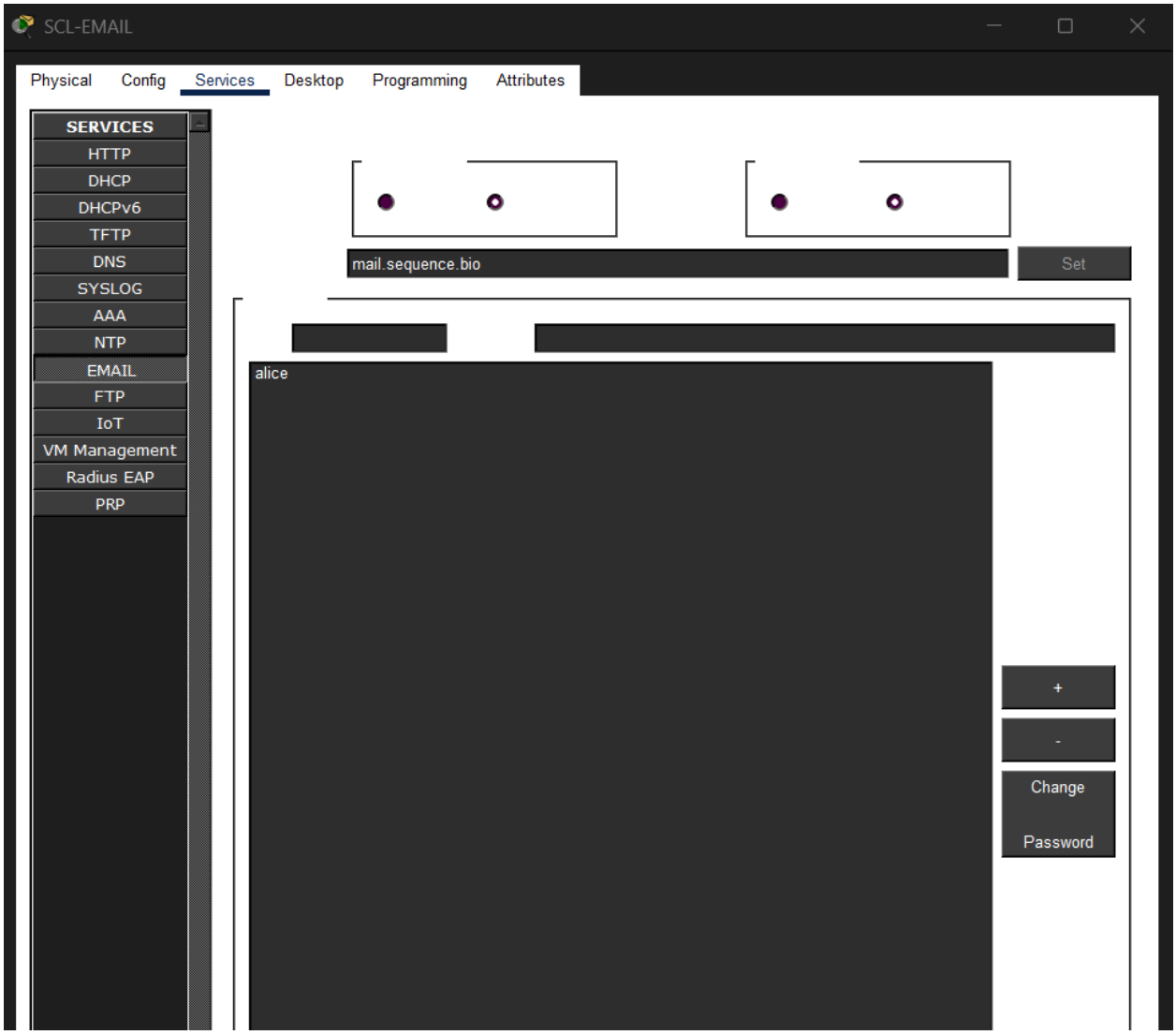

## Web (browser loading the homepage on SCL-PC1)-



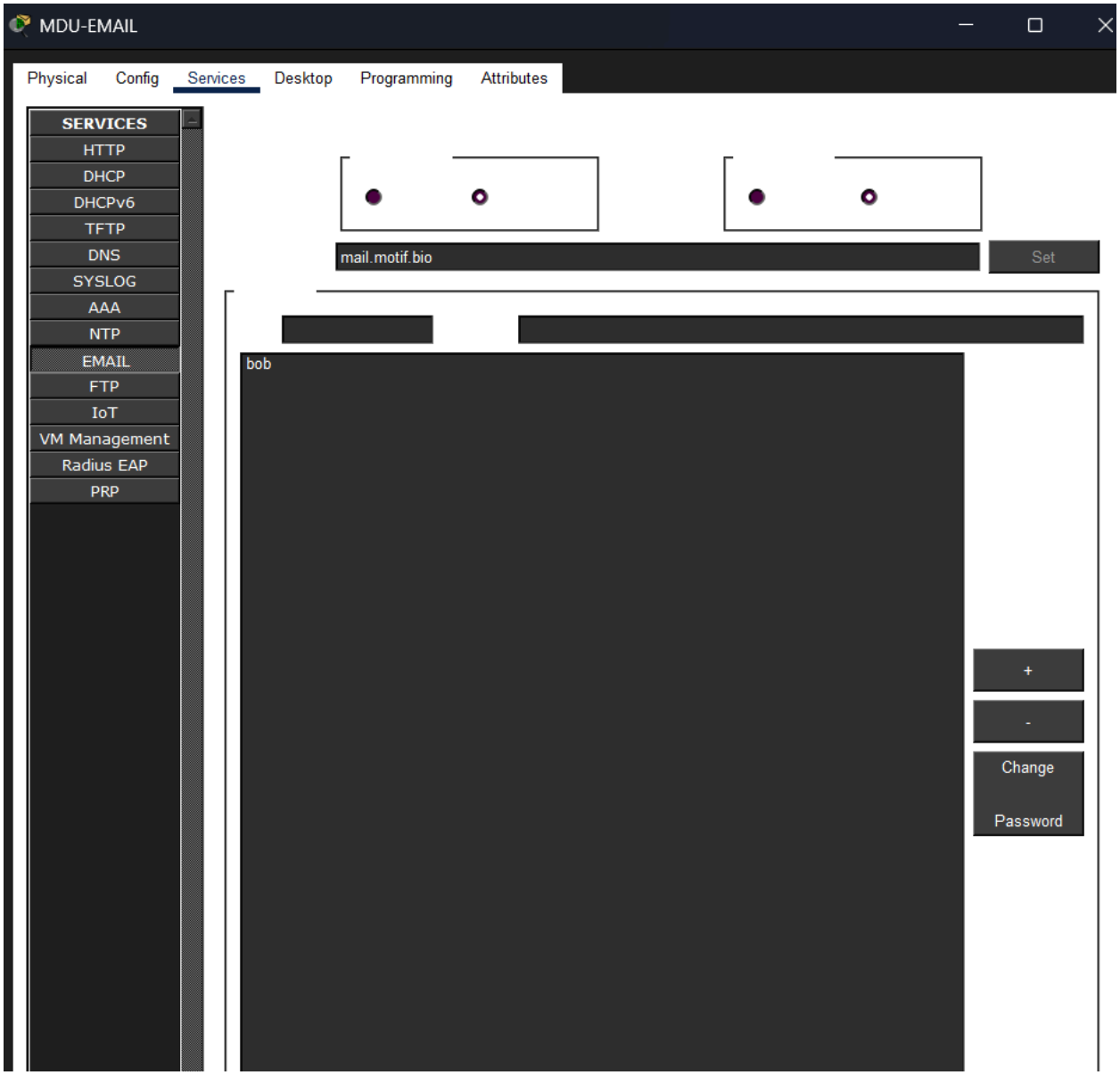

## DHCP (SCL-DHCP — server-based pools)-



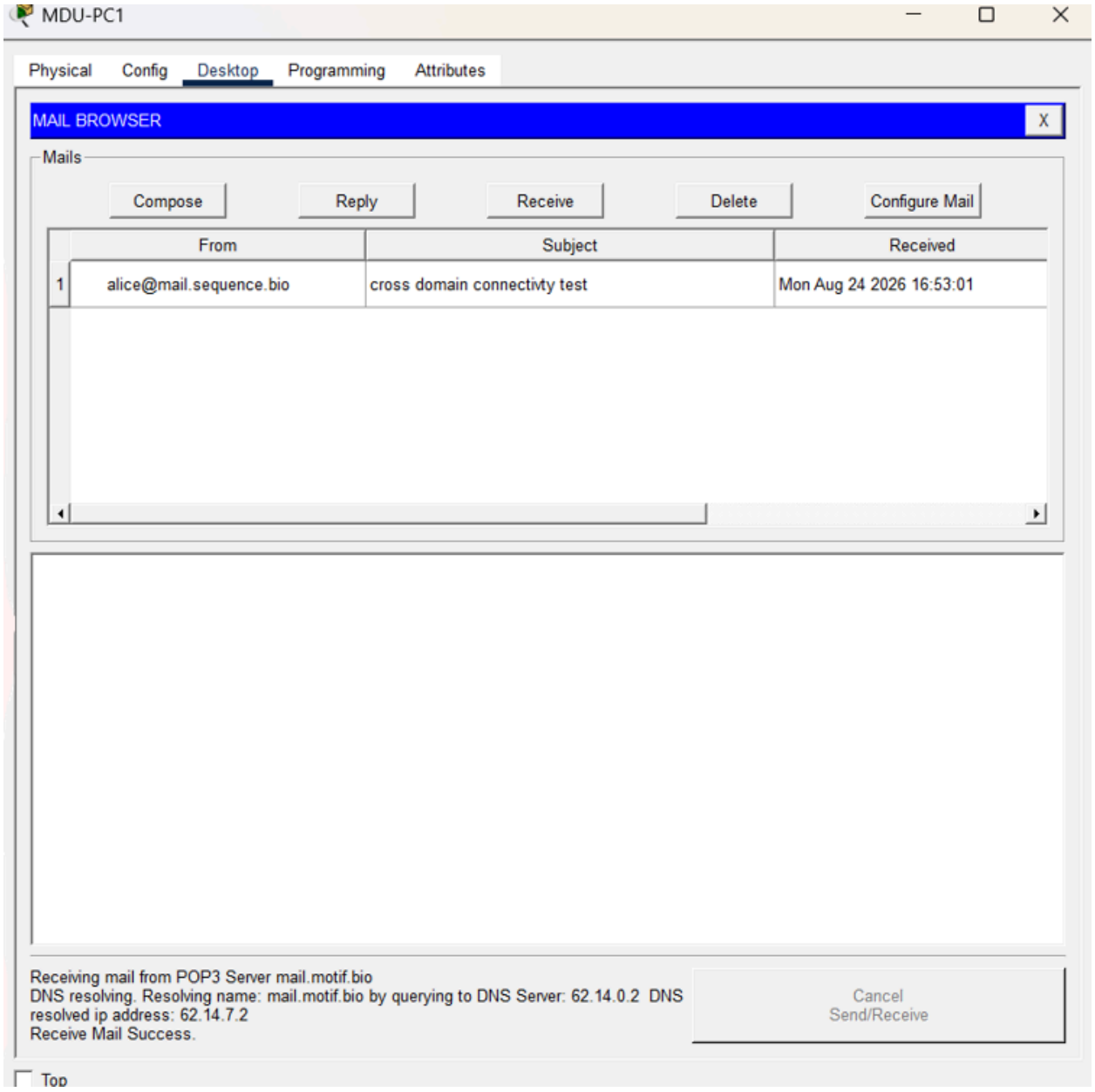

# Email (SCL-EMAIL)-



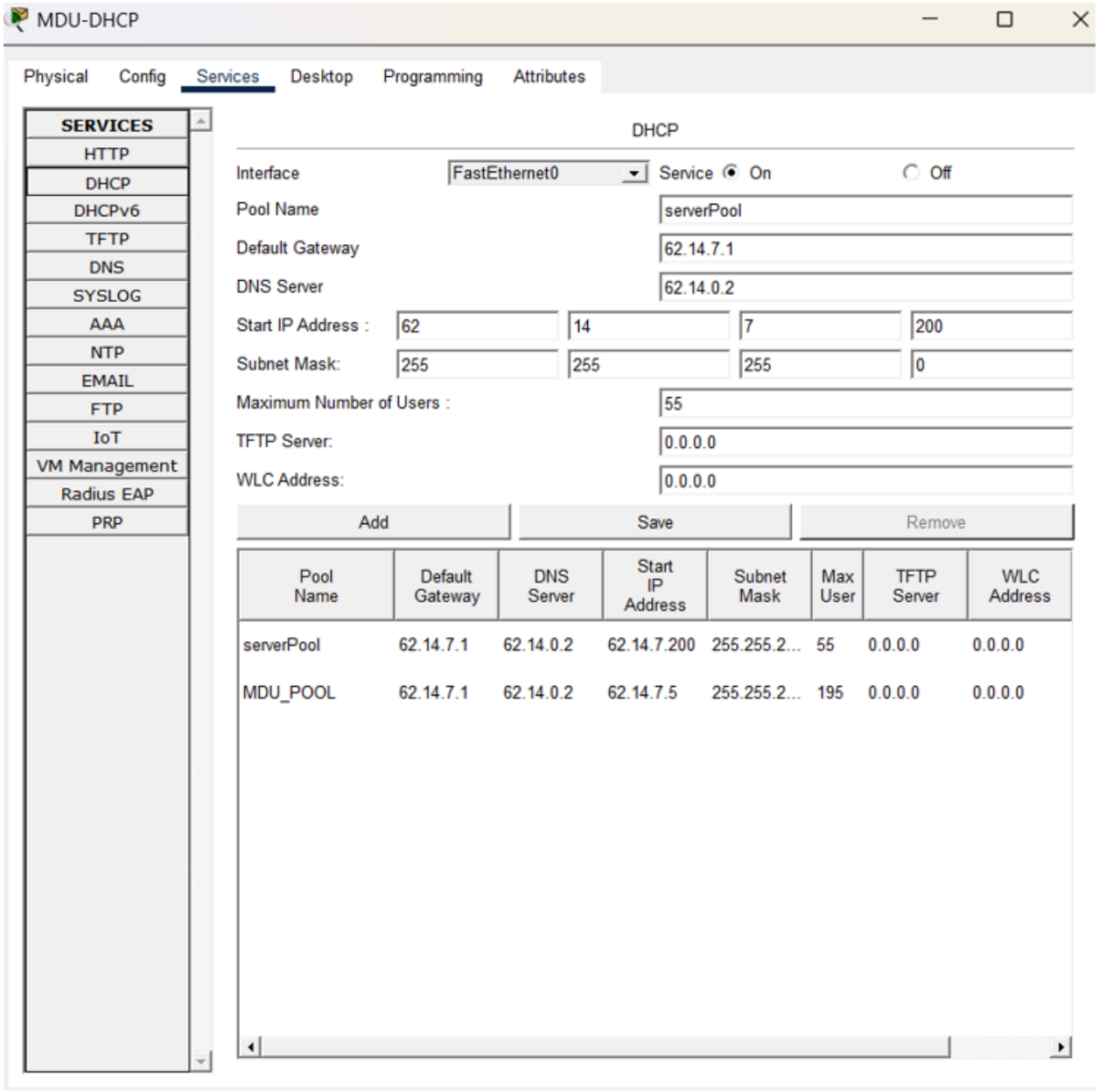

# Email (MDU-EMAIL)-




# MAIL SUCCESS-


MDU-PC1
Physical
Config
Desktop
Programming
Attributes
MAIL BROWSER
X
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
1
alice@mail.sequence.bio
cross domain connectivty test
Mon Aug 24 2026 16:53:01
Receiving mail from POP3 Server mail.motif.bio
DNS resolving. Resolving name: mail.motif.bio by querying to DNS Server: 62.14.0.2 DNS resolved ip address: 62.14.7.2
Receive Mail Success.
Cancel
Send/Receive
Top

# MDU-DHCP-




# 5. Assumptions and Design Decisions

1. The base network is 62.14.0.0/16, derived from the last two digits used by the first two group members (62 and 14).
2. VLSM is applied from the /16 address space, with the largest LAN host requirements allocated first, followed by smaller LAN and point-to-point subnets.
3. Cisco 2911 routers are used throughout the Packet Tracer topology.
4. The RMU/DBG/MDU transit segment uses the shared 62.14.8.192/29 backbone subnet. The remaining inter-router point-to-point connections use /30 subnets.

5. /30 subnets are used for point-to-point connections because each such link requires two usable IP addresses.
6. The first usable address (.1) is used as the default gateway for each LAN. Static servers and printers are assigned low host addresses as shown in the addressing table.
7. ASN is intentionally configured as a fully static network. Its printer and two PCs use static addressing, and ASN does not participate in RIPv2.
8. SCL and MDU use dedicated server-based DHCP, RMU and DBG use router-based DHCP, and GVU uses DHCP relay to the SCL DHCP server.
9. SCL-DHCP provides the SCL and GVU server-based DHCP pools. Because Packet Tracer's protected serverPool could not be removed, the SCL /22 address space is divided into two valid, non-overlapping DHCP ranges.
10. DNS service is provided by SCL-DNS at 62.14.0.2, which is used as the DNS server address for the configured DHCP pools and static hosts.
11. The web service is implemented and tested using Packet Tracer's HTTP server/browser functionality. HTTPS is outside the scope of this project unless explicitly configured.
12. Email service is provided by the configured SCL and MDU email servers, with successful email delivery demonstrated in the project.
13. Unused router interfaces are left without IP addressing and/or are shutdown where appropriate. Only interfaces required by the topology are active.
14. RIPv2 is used by SCL, RMU, DBG, MDU, and GVU, with automatic summarization disabled. LAN-facing interfaces are configured as passive interfaces.
15. GVU uses both static and dynamic routing. It has a static route to the ASN LAN and redistributes that static route into RIP so that the other RIP routers can learn the ASN network.
16. The DBG–MDU /30 connection is treated as a backup path. The floating static route uses administrative distance 123, so the normal RIP route with administrative distance 120 remains preferred while available.
17. The supplied IP addressing table and final Packet Tracer configuration are treated as the authoritative implementation of the proposed network design; the documentation reflects the final configured values.